DO NOW — RECALL

Software development methods: evaluation

Name: _____ Class: _____ Date: _____

1. Independent tasks take 3 and 6 min; final integration 2 min. What is ideal parallel duration?

2. Name one XP practice intended to find code errors early.

3. Why would a user evaluate a prototype?

4. Simplify F=(¬A ∧ ¬B) ∨ (¬A ∧ B). State the law and final answer.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

max(3,6)+2=8 minutes.

2. Model answer

Test-driven development or continuous integration.

3. Model answer

To identify missing needs, usability issues or changes before more costly development.

4. Model answer

Factor ¬A: F=¬A∧(¬B∨B)=¬A∧1=¬A. This also forms a pair of adjacent 1s on a K-map.

Vocabulary — know these terms

evaluation — Judgement supported by evidence and limitations.

risk-driven — Development priorities guided by significant uncertainties.

iterative — Revisits improvements in cycles.

stakeholder — Person or organisation affected by a project.

NEW LEARNING

Read, annotate and explain

Avoid a universal answer

Method suitability depends on risk, requirements, regulation, user access and timescale.

Support with scenario evidence

A high-risk system may justify Spiral's explicit risk analysis despite extra overhead.

Include a drawback

Evaluate cost, documentation, flexibility and technical quality rather than merely naming a method.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

A safety-critical device uses unfamiliar hardware. Evaluate Spiral versus RAD.

Predict the method

Recommend Spiral because technical risks can be evaluated at each iteration.

Check and explain

Explain the disadvantage: risk analysis adds project management and cost.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Give one defining feature each of Waterfall, Spiral, RAD and XP.

Question 2

Choose a development method for a regulated safety-critical control application with uncertain technology. Give three reasons and one drawback.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Evaluate Agile compared with Waterfall for a government reporting system where requirements are partly fixed but user interface needs may change.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

The best development methodology is the same for every software project.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.